DATA SCIENCE • MACHINE LEARNING • RETAIL ANALYTICS • 2026

Anél Henning

DATA SCIENTIST | RETAIL ANALYTICS

C. 813.743.1661
Tampa, FL | Open to Remote
MS Cybersecurity — Purdue Global University (2026)
BS Computer Science | BS Graphic Design (MICA)

CORE SKILLS

Python • R Studio • SQL • Excel (Pivot, Power Query)
Machine Learning • Statistical Analysis • ANOVA
Customer Segmentation • Time Series Forecasting
Power BI / Tableau • Data Storytelling • KPI Dashboards
Workflow Documentation • Business Intelligence

PORTFOLIO PROJECTS

01 — Customer Segmentation (K-Means Clustering)
02 — Sales Forecasting (Time Series / ARIMA)
03 — Product Recommendation System (Collaborative Filtering)
04 — Retail KPI Dashboard (Power BI / Excel)

Customer Segmentation

Unsupervised Learning • K-Means Clustering • Retail Personalization

Python | scikit-learn | K-Means | Pandas | Matplotlib | Seaborn

Business Problem

A grocery retailer needs to understand which customers share purchasing behaviors so marketing teams can deliver targeted promotions, reduce churn, and increase basket size. Without segmentation, one-size-fits-all campaigns underperform across diverse shopper types.

Approach

Applied K-Means clustering (k=4, elbow method) to transactional data covering purchase frequency, average spend, category mix, and recency (RFM model). Feature engineering reduced 40+ raw fields to 6 normalized dimensions. Silhouette score validated cluster separation before labeling segments for business use.

Key Findings

- Identified 4 distinct customer segments: Budget Loyalists, Premium Fresh Shoppers, Convenience Seekers, and Health-Focused Buyers.
- Premium Fresh Shoppers (18% of base) drove 34% of revenue — highest LTV cluster.
- Convenience Seekers showed highest churn risk; targeted coupon campaign modeled to recover ~12% with minimal discount depth.
- Cluster profiles fed directly into a personalized email recommendation engine prototype.
- Academic foundation: one-way ANOVA analysis (Purdue, Foundation of Data Analysis) validated group differences in spend across segments.

4	34%	87%	6
Customer Segments	Revenue from Top Cluster	Silhouette Score	RFM Features

Tools & Methods

Python (scikit-learn, pandas, matplotlib, seaborn) • RFM feature engineering • Elbow method + Silhouette analysis • K-Means (k=4) • Bar charts & scatter plots for stakeholder presentation • ANOVA validation of cluster distinctiveness (R Studio)

Academic Link

Extends coursework from Purdue Global ITP-527: Foundation of Data Analysis — where one-way ANOVA was applied to worldwide box office revenue grouped by MPAA rating, demonstrating statistically significant differences between groups ($p < 0.05$).

Sales Forecasting

Time Series Analysis • ARIMA • Retail Demand Planning

Python | statsmodels | ARIMA | Prophet | Pandas | Excel

Business Problem

Retail buyers and store managers need accurate weekly sales forecasts to optimize inventory replenishment, reduce out-of-stocks, and cut overstock shrink. Poor forecasting directly erodes margin in perishable-heavy departments like produce and deli.

Approach

Built a time series forecasting pipeline on 3 years of weekly sales data across 12 product categories. Applied ARIMA (p=1,d=1,q=1) and compared against Facebook Prophet to capture trend, seasonality, and holiday effects. Evaluated with RMSE and MAPE on a held-out 12-week test set. Final model integrated a promotional calendar as an exogenous variable.

Key Findings

- Prophet outperformed ARIMA on produce categories (MAPE 8.2% vs 11.4%) due to strong seasonal patterns and holiday spikes.
- Holiday weeks (Thanksgiving, Easter) required manual override regressors — without them, forecast error doubled.
- Forecasts reduced projected overstock by ~15% across perishable SKUs when piloted in simulation.
- Weekly forecast dashboard built in Excel (Power Query + dynamic charts) for buyer review.
- Academic foundation: TRUCKS Transportation route and fuel optimization analytics project (Purdue, Statistics for IT) applied similar data-driven decision frameworks.

8.2%	12 wks	3 yrs	15%
Best MAPE (Prophet)	Forecast Horizon	Training Data	Overstock Reduction

Tools & Methods

Python (statsmodels, pandas, matplotlib) • Facebook Prophet • ARIMA / SARIMA • Excel Power Query + dynamic charts • Promotional calendar exogenous variables • RMSE + MAPE evaluation • Train/test split (80/20 temporal)

Academic Link

Built on work from Purdue Global ITP-517: Statistics for IT — stakeholder presentation analyzing TRUCKS Transportation route optimization, fuel efficiency monitoring, and load management using Excel and PowerPoint. Demonstrates applied statistical thinking in a real operations context.

Product Recommendation System

Collaborative Filtering • Content-Based • Personalization

Python | scikit-learn | Surprise | Cosine Similarity | SQL | Pandas

Business Problem

Grocery shoppers often miss products aligned with their health goals and past purchase patterns. A recommendation engine increases basket size, drives trial of new SKUs, and builds personalized loyalty — directly supporting a natural/organic retailer's brand promise.

Approach

Built a hybrid recommendation system combining user-based collaborative filtering (Pearson similarity on purchase history) with content-based filtering on product attributes (category, dietary tags: organic, vegan, gluten-free). Used the Surprise library for matrix factorization (SVD). Evaluated with Precision@5 and Recall@5 on a 20% holdout. SQL queries handled item-user interaction matrix extraction.

Key Findings

- Hybrid model (70% collaborative + 30% content) outperformed pure collaborative filtering by 9 points on Precision@5 (0.61 vs 0.52).
- Cold-start users (< 3 purchases) served via content-based fallback using dietary preference survey at signup.
- Top recommended categories for Health-Focused segment: supplements, organic produce, plant-based proteins — aligned with Customer Segmentation project findings.
- Proof-of-concept API endpoint built to serve top-5 recommendations per user_id.
- Demonstrates machine learning model deployment pipeline suitable for production use.

0.61	0.58	SVD	Hybrid
Precision@5	Recall@5	Matrix Factorization	Model Type

Tools & Methods

Python (scikit-learn, Surprise SVD, pandas, numpy) • Collaborative filtering (Pearson similarity) • Content-based filtering (cosine similarity) • SQL for interaction matrix extraction • Precision@K / Recall@K evaluation • REST API prototype for recommendation serving

Connection to Segmentation

Recommendation outputs fed directly from the 4 customer segments identified in Project 01 — ensuring each segment receives category-appropriate suggestions. Premium Fresh Shoppers receive organic upgrade nudges; Budget Loyalists receive bulk-buy and store-brand alternatives.

Retail KPI Dashboard

Business Intelligence • Power BI / Excel • Executive Reporting

Power BI | Excel | DAX | Power Query | SQL | Data Storytelling

Business Problem

Retail leadership needs a single source of truth for store performance — sales vs. target, gross margin by department, shrink rate, and customer transaction trends. Scattered reports in email and spreadsheets create lag and misalignment across buying, operations, and marketing teams.

Approach

Designed an executive-level retail KPI dashboard in Power BI with four report pages: (1) Sales Overview, (2) Department Deep Dive, (3) Customer Trends, and (4) Inventory & Shrink. Data model built in Power Query connecting POS, inventory, and customer transaction tables. DAX measures calculated rolling 4-week averages, YoY growth, and shrink % against shrink budget.

Dashboard Pages & KPIs

- Sales Overview: Net Sales vs. Budget, Week-over-Week trend, Top 10 categories by revenue.
- Department Deep Dive: Gross Margin %, Sales/Sq Ft, Units Sold by department with drill-through.
- Customer Trends: Transaction count, Average basket size, New vs. Returning customers, Segment distribution (feeds from Customer Segmentation project).
- Inventory & Shrink: Shrink $ and %, Days of Supply by category, Out-of-stock alerts.
- Mobile-optimized layout for store managers; executive summary PDF export automated weekly.

4	12	Live	2
Report Pages	KPI Metrics	Data Refresh	Audiences

Tools & Methods

Power BI (DAX, Power Query, drill-through, bookmarks) • Excel (Pivot Tables, Power Query, dynamic charts) • SQL for source data extraction • Data model: star schema with fact sales table + dim product, dim store, dim date • Stakeholder presentation design informed by MICA Graphic Design + Purdue statistics coursework

Academic Link

Directly extends stakeholder presentation skills developed in Purdue Global ITP-517 (Statistics for IT — TRUCKS Transportation analytics) and design clarity principles from MICA (BS Graphic Design, 2003). Data storytelling is a core competency applied across all four portfolio projects.

About Anél Henning

Entry-level data scientist with a strong foundation in Python, SQL, R Studio, and statistical analysis. Background spans cybersecurity (Purdue Global, MS in progress), computer science, and graphic design — bringing both analytical rigor and the ability to communicate complex findings clearly to non-technical stakeholders. Experience in retail operations (Publix, 10+ years) provides direct understanding of the business problems these four portfolio projects solve.

Education

2024–2026	**MS, Cybersecurity**	Purdue Global University
Prior	**BS, Computer Science**	Rasmussen College
1999–2003	**BS, Graphic Design**	Maryland Institute College of Art (MICA)

Certifications & Memberships

- HIPAA Business Associate Certification (2026)
- ISC2 Certified in Cybersecurity (CC) — In Progress
- CompTIA Security+ — In Progress
- ISC2 Tampa Bay Chapter • ISACA West Florida Chapter
- IEEE Women in Engineering • AIGA (2002–Present)

Contact

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